Program : **Diploma in Textile Technology**	
Course Code : **3063**	Course Title: **Yarn Manufacture I**
Semester : **3**	Credits: **4**
Course Category: **Program core**	
Periods per week: **4 (L:3, T:1, P:0)**	Periods per semester: **60**

Course Objectives:

- To impart the knowledge of constructional features and working of Spinning preparatory machines
- Know various operations carried out at different stages of pre-spinning processes for converting fiber to combed sliver
- To estimate the draft, production and efficiency of the machines involved.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Basic mathematics		Mathematics I , II	1 , 2
Knowledge of mechanical elements in textile machines		Basic Textile mechanics lab	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Discuss the concepts of Ginning and Mixing, Study the working of Blow room machines.	16	Applying
CO2	Describe the concept and working of carding and drawing machine.	15	Applying
CO3	Discuss the method of lap preparation for combing machine, the working mechanism of comber.	12	Applying
CO4	Apply the concept of cleaning efficiency, draft and estimate the production of Scutcher, carding, drawing and comber machine.	15	Applying
	Series Test	2	

CO-PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3	3				
CO2	3	3	3				
CO3	3	3	3				
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Levels
CO1	**Discuss the concepts of Ginning and Mixing, Study the working of Blow room machines.**		
M1.01	Remember fibre properties and Understand the concept of Ginning, mixing and blending and suggest mixing plans for fine, medium and coarser yarn counts.	4	Applying
M1.02	Understand the function of Blow room, types of cleaning points and machinery sequence in modern Blow room line.	4	Understanding
M1.03	Understand the working of different openers, cleaners, Scutcher and understand the functions of ancillary equipment in Blow room line.	5	Understanding
M1.04	Identify and analyze the possible reasons for blow room processing defects and suggest remedies to solve them.	3	Analyzing

Contents:

Objects of ginning - Types of gins for different cotton - objects of mixing - Hand stack mixing principle - Differentiate Mixing and Blending - Preparation of Mixing plan for fine, medium and coarse counts using Indian cottons - Need for opening and cleaning of cotton - Functions of Blow room - Types of feeding to beaters, major and minor cleaning points - classification of blow room machines - bale opening, mixing, coarse cleaning and fine cleaning machines - Sequence of machines in Modern Blow room Line for processing cotton (Reiter) - Working of MBO - Blendomat - Mono cylinder - Multimixer - ERM cleaner - Two way distributor - Hopper feeder - krischner beater - scutcher - Objectives of Ancillary equipments - metal detector, fire detector, contamination detector, AWES - Speed and setting in Blow room machines, Identify and analyze the possible reasons for blow room processing defects such as Conical lap, Nep formation, Poor cleaning Efficiency, Lap licking, High weight variations of lap and suggest remedies to solve them.

<table>
<tr><td>CO2</td><td colspan="3">Describe the concept and working of carding and drawing machine.</td></tr>
<tr><td>M2.01</td><td>Understand the working of chute feed system and carding machine and study the function of various parts of carding machine.</td><td>4</td><td>Understanding</td></tr>
<tr><td>M2.02</td><td>Identify and analyze the possible reasons for Carding processing defects and suggest remedies to solve them</td><td>4</td><td>Analyzing</td></tr>
<tr><td>M2.03</td><td>Understand the working of Draw frame machine and study the function of various parts of Draw frame machine.</td><td>4</td><td>Understanding</td></tr>
<tr><td>M2.04</td><td>Identify and analyze the possible reasons for Draw frame processing defects and suggest remedies to solve them.</td><td>3</td><td>Analyzing</td></tr>
<tr><td></td><td>Series - test I</td><td>1</td><td></td></tr>
<tr><td colspan="4">Contents:
Working of Chute feed machine (LA7/6) - advantages of chute feed over lap feed - objects of carding - principle of stripping and carding - working of modern Revolving flat carding machine - Functions of Licker-in region, flats, cylinder region, Pre carding and post carding stationary Flats, doffer, web doffing mechanism - crush roller - Diameter and working speed of important parts of card - Working of Long term Auto leveller in card - open loop and closed loop system - objectives of flexible bend - heel and toe arrangement - Card clothing specification for cotton - fibre hooks - types - Suggest Card setting for processing cotton fibre for medium counts - Types of waste and approximate percentage in high production carding, Identify the possible reasons for Carding processing defects such as High Nep count, Poor cleaning Efficiency, Sliver hank variations and suggest remedies to solve them.
Objects of draw frame, working of draw frame, Principle of roller drafting, top roller, bottom roller and weighting system (spring & pneumatic), working of polar drafting and platts (3 over 3) drafting system, Need for auto leveller in drawframe. Identify the possible reasons for Draw frame processing defects such as Roller lapping, Sliver hank variations and suggest remedies to solve them.</td></tr>
<tr><td>CO3</td><td colspan="3">Discuss the method of lap preparation for combing machine, the working mechanism of comber.</td></tr>
<tr><td>M3.01</td><td>Understand the methods of lap preparation for combing in Sliver lap and Ribbon lap machines.</td><td>3</td><td>Understanding</td></tr>
<tr><td>M3.02</td><td>Understand the working of Super lap machine</td><td>3</td><td>Understanding</td></tr>
<tr><td>M3.03</td><td>Understand the working of Comber machine and study the influence of comber setting on noil percentage.</td><td>4</td><td>Applying</td></tr>
</table>

<table>
<tr><td>M3.04</td><td>Understand the process parameters and the influence of comber setting on noil percentage.</td><td>2</td><td>Analyzing</td></tr>
<tr><td colspan="4">Contents:
Method of lap preparation - Conventional method (batt doubling) and modern method (sliver doubling) - need for even number of passage between carding and comber - Objectives of sliver lap, Ribbon lap machine - Objects and working of Super Lap machine, objectives and working of modern comber machine with combing cycle - function of major parts - Types of feed and factors influencing noil %, Brief study about degree of combing - scratch combing, semi combing ,regular combing, Double combing, importance of pre comber draft, Important settings in comber, importance of post comber drawing. Identify the possible reasons for Comber processing defects such as Longer fibres in noil, Short fibres in sliver, Cutting across, Detaching roller lapping and suggest remedies to solve them.</td></tr>
<tr><td>CO4</td><td colspan="3">Apply the concept of cleaning efficiency, draft and estimate the production of Scutcher, carding, drawing and comber machine.</td></tr>
<tr><td>M4.01</td><td>Estimate the cleaning efficiency of blow room machines, hank of lap, production and machine efficiency of scutcher</td><td>4</td><td>Applying</td></tr>
<tr><td>M4.02</td><td>Determine the sliver hank, actual draft, mechanical draft, production and efficiency of carding machine.</td><td>4</td><td>Applying</td></tr>
<tr><td>M4.03</td><td>Estimate the sliver hank, draft, production and efficiency of drawframe machine</td><td>4</td><td>Applying</td></tr>
<tr><td>M4.04</td><td>Determine the production and efficiency of comber machine</td><td>3</td><td>Applying</td></tr>
<tr><td></td><td>Series - Test II</td><td>1</td><td></td></tr>
<tr><td colspan="4">Contents:
Estimation of Blow room cleaning efficiency and combined cleaning efficiency, Estimation of production and efficiency of scutcher, beats per inch of bladed beater, hank calculation of lap and sliver, difference between carding actual draft and mechanical draft, Understand the relation between actual and mechanical draft, Estimation of actual production and efficiency of card, actual draft of draw frame, production and efficiency of draw frame, production of comber.</td></tr>
</table>

Text / Reference:

T/R	Book Title/Author
T1	Oxtoby E., “Spun Yarn Technology”, Butterworths, London,
R2	Werner Klein,“The Rieter Manual of Spinning, Volume.2 - Blow room& Carding”
R3	Werner Klein,“The Rieter Manual of Spinning, Volume.3 - Spinning Preparation”,
R4	R.Chattopadhyay- Advances in Technology of Yarn production-NCUTE
R5	Klein W., “A Practical Guide to Opening and Carding “, The Textile Institute, Carl A. Lawrence., “Fundamentals of Spun Yarn Technology”,

Online resources:

Sl.No	Website Link
1	https://nptel.ac.in
2	www.reiter.com
3	www.lmwtmd.com
4	www.fibre2fashion.com
5	www.textilelearner.blogspot.com